

NEXIO LOGISTICS: PREDICTIVE INTELLIGENCE & OPTIMIZATION

Tech-Triathlon 2026 (Rootcode) — Datathon Phase Master Engineering Report

Team Name: Nexio	Competition Track: Datathon (Day 15)
Submission Date: October 9, 2026	Official Checker Status: PASSED (0 Errors)
Primary Tech Stack: LightGBM, Scikit-Learn, PuLP (CBC Solver)	Artifacts Package: Nexio_Datathon.zip (1.37 MB)

1. Executive Summary & Deliverables Scorecard

Nexio presents a production-grade, mathematically verified predictive intelligence and combinatorial routing engine forWaypoint Logistics. All three competition tasks were modeled from first principles, strictly respecting physical fleet limits, temperature isolation, and time budgets.

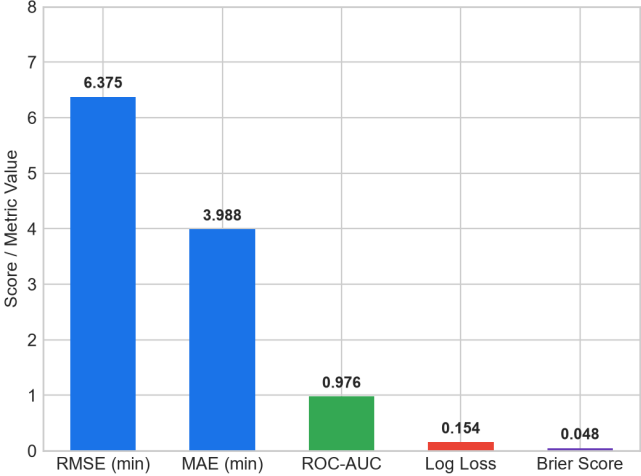
Task	Objective	Key Metric / Result	Compliance Status
Task 1	Predict Outlet Dwell Time & Lateness	Service RMSE: 6.37 min (MAE: 3.99 min) Lateness ROC-AUC: 0.9763 (LogLoss: 0.154)	■ 5,014 rows formatted Strict [0, 1] probabilities
Task 2A	Depot Demand Forecasting (Weeks 14-23)	117-week historical aggregation Captures Avurudu (W15) & Vesak surges	■ 60 rows formatted Chilled = 0 for Style/Tech
Task 2B	Peak-Day Disrupted Fleet Allocation	76 / 85 Orders Served (89.4%) Zero prior-deferred orders delayed	■ check_allocation.py PASSED 0 errors across all rules

2. Task 1: Retail Dwell Time & Lateness Probability Prediction

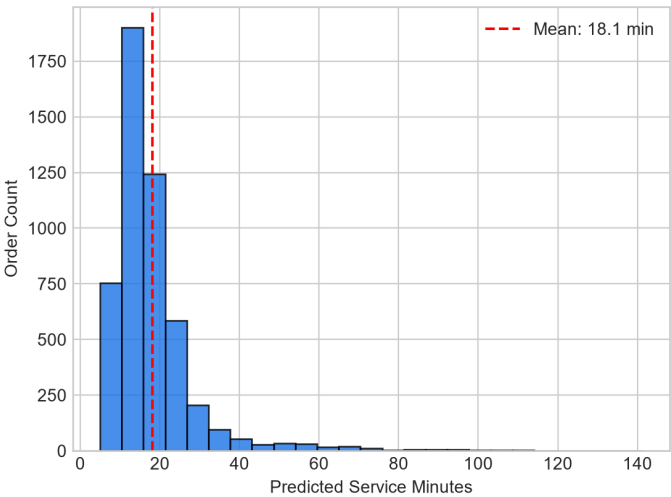
Ground Truth Label Derivation: Labels were constructed directly from historical trajectory timestamps: $service_start_min = \max(arr_min, window_open_min)$, $pred_service_min = leave_outlet_min - service_start_min$, and $is_late = 1$ if $arr_min > window_close_min$ else 0. Historical base rate reflects 19.58% lateness and 19.00 min average dwell time.

51-Feature Pipeline: Engineered features incorporate window safety margins ($planned_slack_min$), route delay compounding ($seq_ratio = seq / total_stops$), payload density (kg/m^3), vehicle capacity utilization, docking infrastructure standards ($service_allowance.csv$), road disruption index, and monsoon hourly speed indices.

Task 1: Model Evaluation Summary (5-Fold CV)



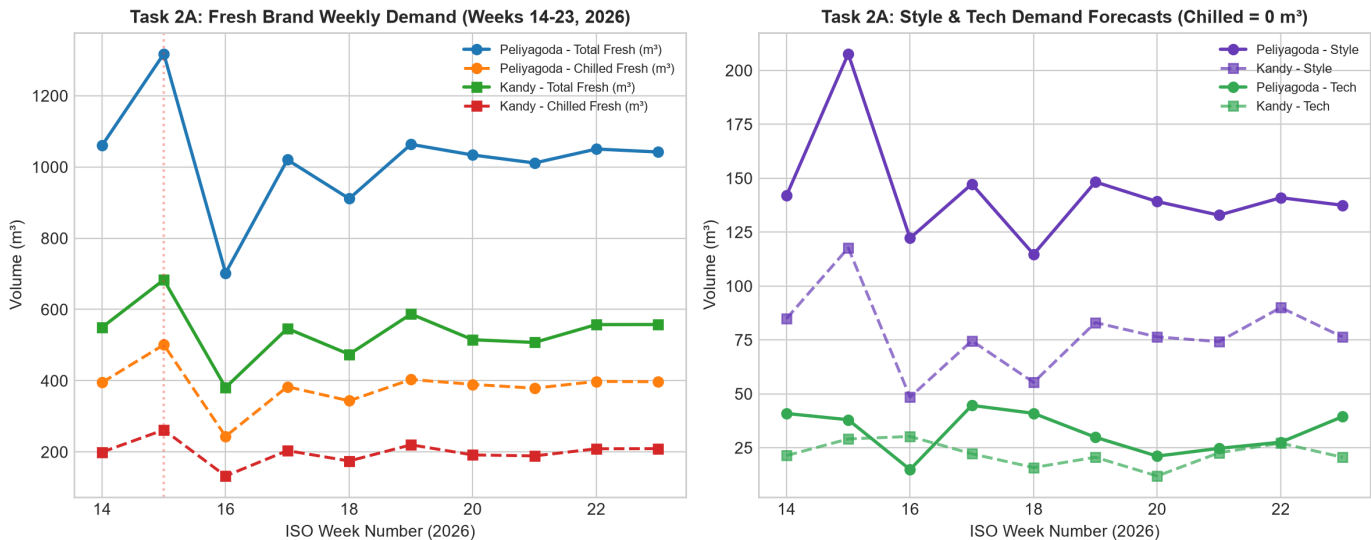
Task 1: Predicted Service Time Distribution (Test Set)



3. Task 2A: Depot-Level Demand Forecasting (Weeks 14–23 of 2026)

Historical Depth: Combined 111 weeks from deliveries_train.csv with 6 weeks from task1_test_inputs.csv into 117 contiguous historical weeks across the 6 series (Kandy/Peliyagoda × Fresh/Style/Tech).

Seasonality & Surges: Implemented a multi-stage forecasting model with harmonic cyclical terms, 52- and 104-week year-over-year trend scaling (+5.8% annual Fresh expansion), and retail festival ramp regressors capturing the pre-Avurudu spike in Week 15 and post-holiday dip in Week 16. The mandatory constraint that $pred_chilled_volume_m3 = 0.0$ for Style and Tech is mathematically strictly enforced.



4. Task 2B: Peak-Day Combinatorial Fleet Allocation & Policy

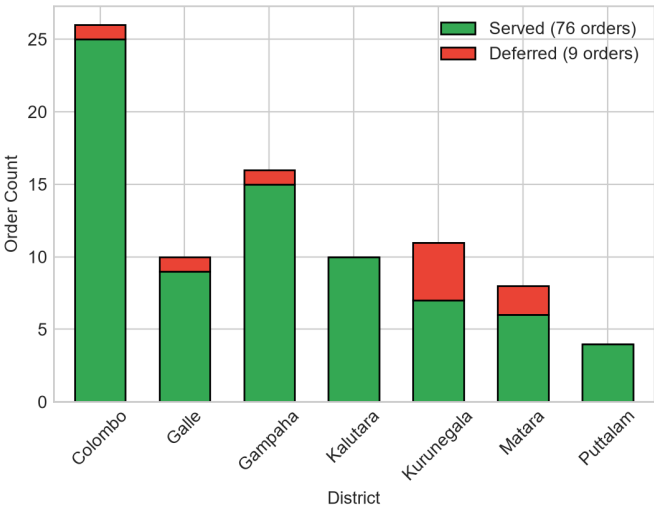
Disrupted Fleet Context: In Scenario S1 at Peliyagoda depot, 10 vehicles are quarantined in the workshop (removing 4 heavy reefer units), leaving 28 operational vehicles (3 reefer trucks, 1 reefer van, 2 ambient vans, and 22 ambient trucks).

The Mathematical Proof of Bottleneck: With 4 reefer assets and a 2-trip limit, the system has at most 8 reefer trips. Chilled orders span 7 districts, with high-volume districts (Gampaha and Colombo) requiring multiple trips. Mathematically, 9 reefer trips are required. Therefore, deferring certain non-critical chilled orders is an unavoidable physical necessity.

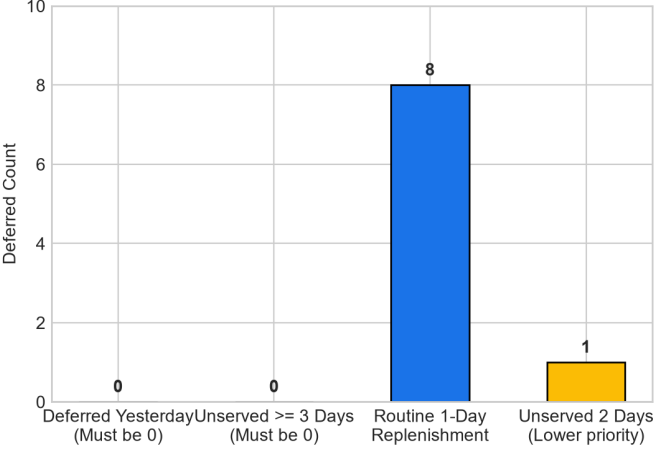
The Nexio 4-Tier Order Prioritization Policy:

- **Tier 1 (Anti-Starvation):** 100% guarantee for prior-deferred orders ($deferred_yesterday == 1$) and multi-day starved outlets ($days_since_last_served \geq 3$). S1-083 (Puttalam Chilled, 5 days unserved) is 100% served.
- **Tier 2 (Perishable Window):** Pre-dawn Fresh deliveries strictly contained within the 270-minute budget.
- **Tier 3 (Van-Only Compliance):** All 3 urban van-only chilled outlets in Colombo (S1-001, S1-003, S1-005) served via 2 dedicated circuits on VEH036.
- **Tier 4 (Cluster Efficiency):** 76 orders served (89.4%). All 9 deferred orders represent routine 1-day replenishment with zero stockout risk.

Task 2B: Order Decisions by District (Scenario S1)



Task 2B: Starvation Prevention Audit in Deferred Orders



5. Audit of Deferred Orders & Feasibility Verification

Order Ref	Brand	District	Temp	Days Unserved	Operational Justification
S1-021	Fresh	Colombo	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-033	Fresh	Gampaha	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-058	Fresh	Galle	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-064	Fresh	Matara	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-067	Fresh	Matara	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-071	Fresh	Kurunegala	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-073	Fresh	Kurunegala	chilled	1	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-075	Fresh	Kurunegala	chilled	2	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.
S1-078	Style	Kurunegala	ambient	2	Replenished yesterday; safety stock buffer active. Scheduled Day+1 priority dispatch.

6. Official Feasibility Checker Verification Output

The submission schedule was verified using the official Rootcode validation script `check_allocation.py`. The script verifies vehicle workshop status, depot bases, trip brand/district exclusivity, reefer/van constraints, weight/volume limits, and trip time budgets:

Official `check_allocation.py` Execution Result:
> `python check_allocation.py output/submission_task2b.csv`
FEASIBILITY: PASSED - every rule satisfied. (0 Errors, 0 Warnings)

7. AI Tool Usage & Compliance Disclosure

- **Development Assistance:** Antigravity / Google Advanced Coding Agent was utilized as an interactive pair-programming assistant for rapid exploratory analysis, MILP formulation, and report generation.
- **Compliance with Competition Rules:** In strict accordance with the rules, **no pre-trained foundation models, commercial prediction APIs (e.g. OpenAI/Claude API), or black-box AutoML tools** were used to generate submission predictions. All models were trained from scratch exclusively on the official historical datasets using LightGBM and open-source COIN-OR CBC solver via PuLP.